BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 6465

Retyped faithfully from the original board paper — same questions, no changes.

Paper: I Objective-(iii)

بائیولوجی

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6465

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

نوٹ:- ہر سوال کے چار ممکنہ جوابات A, B, C اور D دیئے گئے ہیں۔ جوابی کاپی پر ہر سوال کے سامنے دیئے گئے دائروں میں سے درست جواب کے مطابق دائرہ کو مار کر یا پین سے پُر کیجیے۔ ایک سے زیادہ دائروں کو پُر کرنے یا کاٹ کر پُر کرنے کی صورت میں مذکورہ جواب غلط تصور ہوگا۔

Q1. In most birds the number of air sacs are:

(A) 6 (B) 7

(C) 9 (D) 8

Q2. Which of the given animals is not omnivorous?

(A) Crow (B) Man

(C) Pig (D) Dog

Q3. NADH in respiratory electron transport chain is oxidized by:

(A) FADH (B) O_2

(C) Coenzyme Q (D) Cytochrom[illegible in source scan]

Q4. How many molecules of glyceraldehyde 3-phosphate are required to produce one molecule of glucose?

(A) 2
(B) 1
(C) 3
(D) 6

Q5. Paired gill openings are developed in all chordates but non-functional in:

(A) Rat
(B) Fish
(C) Frog
(D) Amphiox[illegible in source scan]

Q6. A respiratory pigment of blue colour called haemocyanin is present in animals of phylum:

(A) Porifera
(B) Coelenterata
(C) Mollusca
(D) Arthropo[illegible in source scan]

Q7. In which phylum sexual phase is not observed?

(A) Basidiomycota
(B) Deuteromycota
(C) Ascomycota
(D) Zygomyco[illegible in source scan]

Q8. The simplest of all bryophytes are:

(A) Mosses
(B) Hornworts
(C) Whisk ferns
(D) Liverwor[illegible in source scan]

Q9. African sleeping sickness is caused by:

(A) Trypanosoma
(B) Entamoeba
(C) Plasmodium
(D) Stento[illegible in source scan]

Q10. Which of the given are spiral shaped bacteria?

(A) Cocci
(B) Pseudomonas
(C) Bacilli
(D) Vibrio

Q11. Independent evolutionary unit among the given is:

(A) Species
(B) Population
(C) Genus
(D) Family

Q12. Vacuole in plants is responsible for:

(A) Photosynthesis
(B) Cellular excretion
(C) Turgor pressure
(D) Starch sto[illegible in source scan]

Q13. Vitamins are essential raw materials for the synthesis of:

(A) Coenzyme
(B) Prosthetic group
(C) Holoenzyme
(D) Apoenzym[illegible in source scan]

Q14. Carbohydrates are organic molecules and contain three elements:

(A) Carbon, H_2O and oxygen

(B) Carbon, sulphur and hydrogen

(C) Carbon, calcium and hydrogen

(D) Carbon, hyd[illegible in source scan] and oxyg[illegible in source scan]

Q15. Branch of Biology which deals with study of ancestral history of living organisms is called:

(A) Palaeontology

(B) Zoogeography

(C) Evolution

(D) Heredit[illegible in source scan]

Q16. The substance produced by basophils that inhibits blood clotting is:

(A) Heparin

(B) Histamine

(C) Fibrin

(D) Albumi[illegible in source scan]

Q17. Which vein has oxygenated blood?

(A) Femoral vein

(B) Pulmonary vein

(C) Subclavian vein

(D) Jugular v[illegible in source scan]

Paper: I Subjective

بائیولوجی

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

> Note: Note:- Section B is compulsory.

Q2. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) How does fibrous protein differ from globular protein? (2 points required)

(ii) Differentiate the enzyme and coenzyme.

(iii) How does substrate concentration affect the rate of enzyme action?

(iv) Define mycorrhizae. Give one example.

(v) Write down medicinal importance of fungi. (2 points required)

(vi) Give four characteristic features of phylum Annelida.

(vii) Write two benefits and two harms caused by insects.

(viii) Define irreversible inhibitors.

(ix) What are tube feet?

(x) How would you differentiate polyps and medusae?

(xi) Give molecular formula of chlorophyll "a" and "b".

(xii) What are action and absorption spectra?

Q3. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Differentiate the micro and macro molecules.

(ii) Define bioremediation. Give one example.

(iii) Define resolution of human eye.

(iv) What are micro filaments? Give their function.

(v) What is giant amoeba? Give its biological name.

(vi) Differentiate the tests of Foraminifera and Actinopods.

(vii) Define thallus. In which organisms is it found?

(viii) What are diatoms? Give one example.

(ix) Define photorespiration.

(x) What are lenticels? Give their function.

(xi) What is apoplast pathway of water transport in plants?

(xii) Define thrombus and embolus.

Q4. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) Define species.

(ii) Differentiate the structure and function of flagella and pili.

(iii) Write down the scientific name of sugar cane and brinjal.

(iv) Why is sphenopsida called arthrophytes?

(v) What is an embryo sac?

(vi) Define circinate vernation.

(vii) Define nutrition. Enlist its types.

(viii) Food habit can be responsible for heartburn. How?

(ix) Trace the route of amino acids and large fatty acids from digestive tract and blood stream.

> Note: Attempt any THREE questions. Each question carries EIGHT (4+4=8) marks. (8x3=24)

Q5. (Marks 8)

(a) [4] Define cloning. Discuss its mechanism and commercial importance.

(b) [4] What is pulmonary respiration? Discuss its phases in frog.

Q6. (Marks 8)

(a) [4] Write down a note on RNA and its types.

(b) [4] What are the economic gains due to fungi?

Q7. (Marks 8)

(a) [4] What are plastids? Describe different types of plastids.

(b) [4] Describe different mechanical and chemical processes occurring in oral cavity for digestion of food in man.

Q8. (Marks 8)

(a) [4] Give characteristics of viruses.

(b) [4] Explain the mechanism of transpiration pull in cohesion tension theory.

Q9. (Marks 8)

(a) [4] Write down a note on: i. Nucleoid ii. Plasmid

(b) [4] Describe three phases of Calvin cycle.